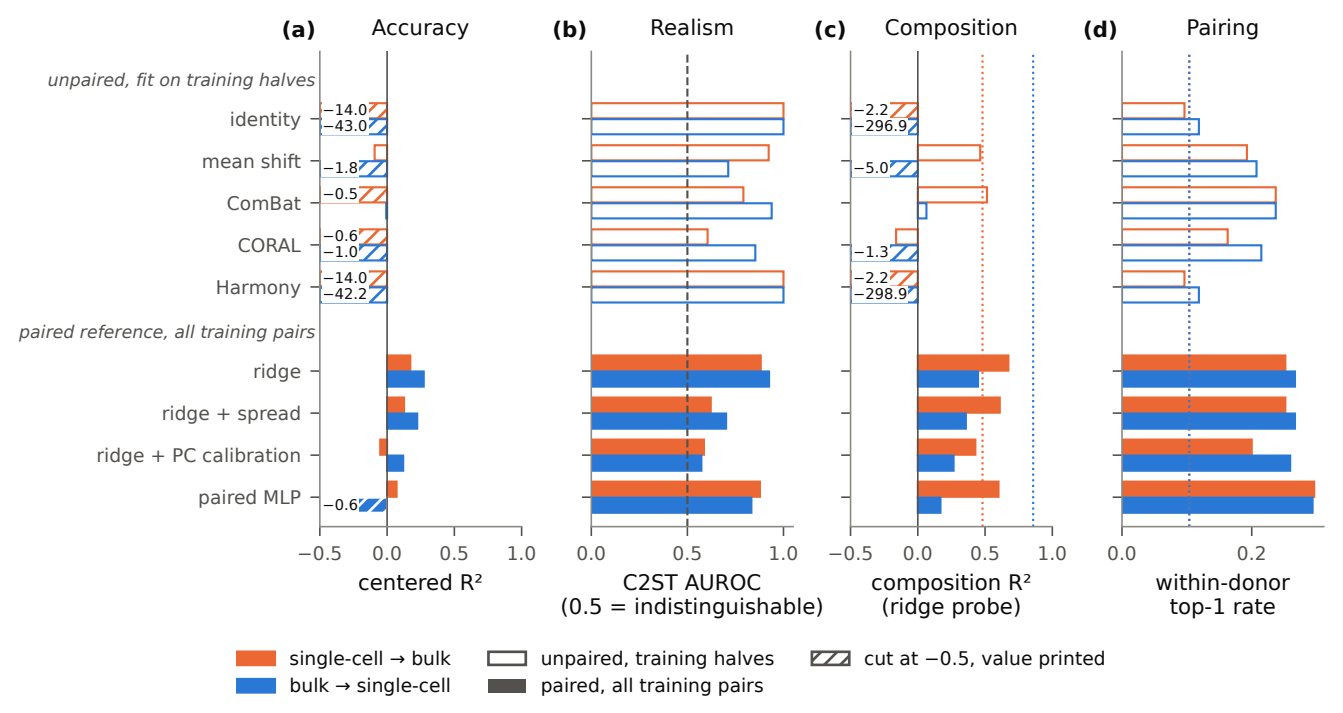
(a) Accuracy
(b) Realism
(c) Composition
(d) Pairing
unpaired, fit on training halves
identity
mean shift
ComBat
CORAL
Harmony
paired reference, all training pairs
ridge
ridge + spread
ridge + PC calibration
paired MLP
−14.0
−43.0
−1.8
−0.5
−0.6
−1.0
−14.0
−42.2
−0.6
−2.2
−296.9
−5.0
−1.3
−2.2
−298.9
−0.5
0.0
0.5
1.0
0.0
0.5
1.0
−0.5
0.0
0.5
1.0
0.0
0.2
centered R²
C2ST AUROC
(0.5 = indistinguishable)
composition R²
(ridge probe)
within-donor
top-1 rate
single-cell → bulk
bulk → single-cell
unpaired, training halves
paired, all training pairs
cut at −0.5, value printed